University of BRISTOL

Introduction and logistics

CADE30008 Flight Dynamics & Advanced Control

Dr. Steve Bullock

Schedule

Week	Act	This week	Coursework	Laboratory
1		Lecture 1: The design cycle, end to end		**Quanser** open access
2		No control session. Flight dynamics has both of the week's slots.		
3		Tuesday · Lecture 2: Requirements and models you can trust		
4	I	Tuesday · Lecture 3: PID, properly		
5		Tuesday · Guest lecture	**Brief released**	
6		Consolidation week · no lecture	**Checkpoint 1, provisional**	
7		Tuesday · Lecture 4: Stability and margins		
	II	Thursday (TBC) · Lecture 5: Robustness and trade-offs		
8		Lecture 6: Loop shaping	**Checkpoint 2, provisional**	
9		Lecture 7: Flight control architecture	**Checkpoint 3, provisional**	
10	III	Lecture 8: State space and state feedback		
11		Lecture 9: What comes next · then Coursework Q&A	**Due Thursday of week 11**	
12		Revision week		

Lecture · Guest lecture · Consolidation · No session · Coursework due Thursday of week 11

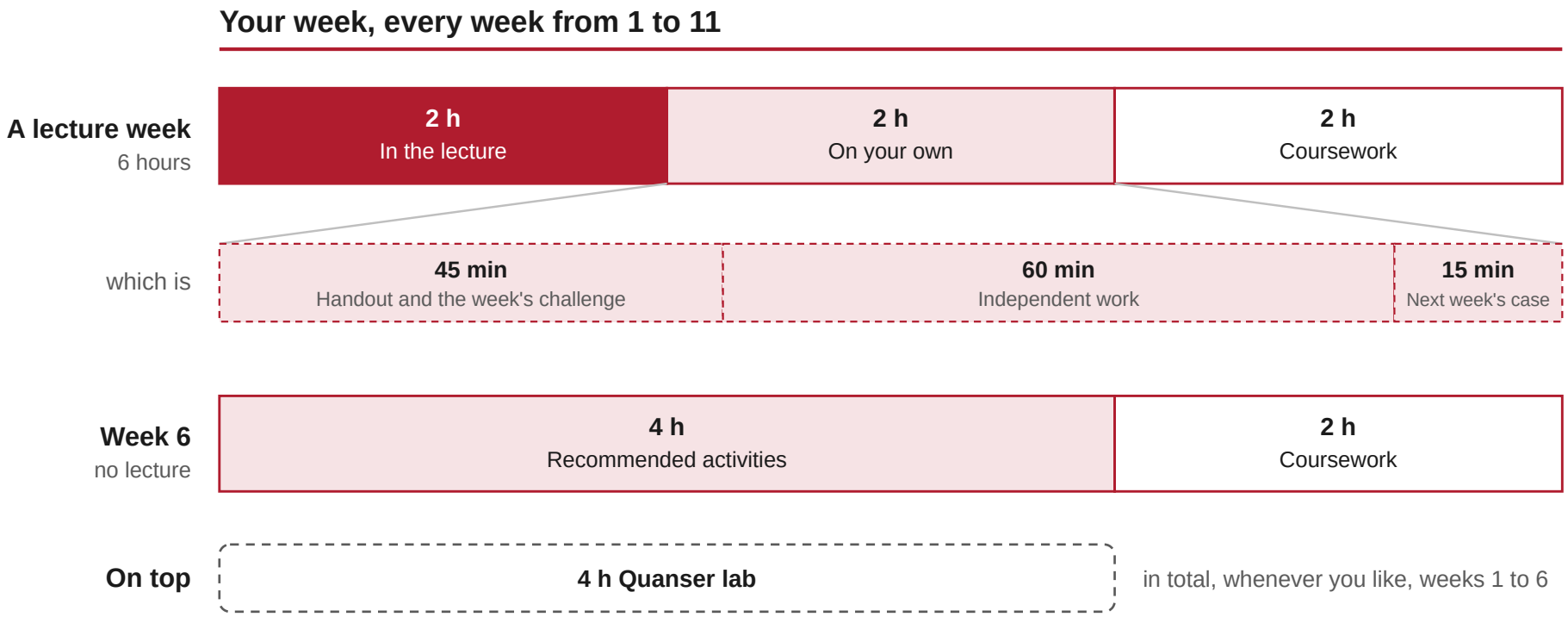
Assessment

- **The coursework counts for 100% of your grade.** One piece, due Thursday of week 11
- Everything else is **formative** — it doesn't contribute to your grade *directly*
- **But it is how the coursework gets built.** The checkpoints *become* the paper
- **Checkpoints, provisionally weeks 6, 8 and 9.** Automatic checks, a message to the whole cohort, and a personal note for some of you
- Nothing formative is randomised. Work on it together if you like

Brief release: week 5

- It goes through **external examiner review** first, which is why it is not out on day one
- **Until then, the best preparation is the unit itself.** What you do in and out of these sessions is what the coursework asks for
- **Front-load the laboratory.** It is open access, slots are booked, and the window closes well before the deadline
- **Watch the videos.** Brian Douglas and Steve Brunton, linked from the reading page

Your week


Your week, every week from 1 to 11
A lecture week
6 hours
2 h
In the lecture
2 h
On your own
2 h
Coursework
which is
45 min
Handout and the week's challenge
60 min
Independent work
15 min
Next week's case
Week 6
no lecture
4 h
Recommended activities
2 h
Coursework
On top
4 h Quanser lab
in total, whenever you like, weeks 1 to 6

The Quanser laboratory



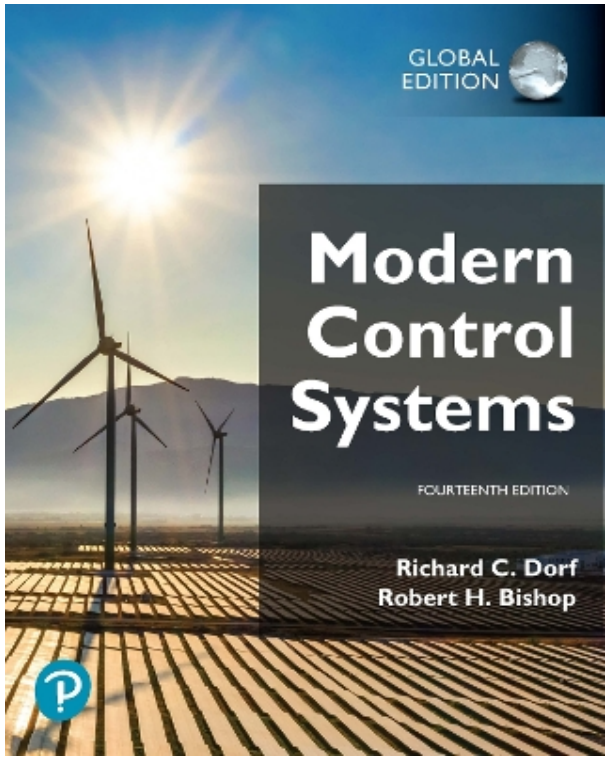
Where M.003, under the Queens Building cafe, with the flight simulators

Booking Link on Blackboard.

Who with Pairs or threes, one of you books

Four hours, open access until end of week 6.

One book worth knowing about



Dorf and Bishop, *Modern Control Systems*
— Pearson, any edition from the 12th

- **You do not have to buy it.** The handouts are the authoritative version and stand on their own
- Closest in spirit to this unit: it frames everything around **the design cycle**
- Use it for **a second explanation** to complement these resources, and for **more worked examples**
- The library has it. Which sections go with which week is on the reading page

Share the library copies

20 print copies and 14 eBook licences, for about 190 of you

The 12th, 13th and 14th editions added together, not copies of one edition.

- Reading an eBook online locks that copy while you have it open.
- Downloading the whole book locks a copy for 24 hours.
- Downloading a PDF chapter locks nothing, but is capped at 56 pages.

Download the chapter you want and get out.

AI in this unit

- **AI rewards expertise.** Anyone can produce passable-looking work, until it comes up against an expert *or the real world*
- **Work, or gym?** If only the result matters it's work. If *doing* it is the point, it's the gym

> A design in a domain you don't understand, produced by a tool you don't understand and can't check, is **not responsible, not ethical, not engineering**.

What you're here to build

Judgement: knowing a plausible answer is wrong, before you can say why.
Taste: knowing which of several correct designs is the good one.

Nobody can hand you either — not me, not a model. Do the hard part.